30-Day Readmission Analysis Among Diabetic Patients

Capstone Project 1 | Data Analytics using Python, MySQL and Power BI

Project report

GitHub: https://github.com/anumasi-analytics/diabetes-readmission-analysis

Executive summary

This project studies which patient and hospital-stay factors are associated with readmission within 30 days among diabetic patients. It uses 101,766 hospital encounters from 130 US hospitals. The data was profiled and cleaned in Python, leaving 69,979 patients (one row per patient). The clean data was stored in MySQL, analysed with 9 SQL queries, and presented in a 3-page Power BI dashboard built on a star-schema data model.

Overall, **8.98%** of patients (6,283 of 69,979) were readmitted within 30 days. The strongest factor is the number of prior visits: 7.94% for patients with none and 18.00% for patients with six or more. Longer stays and older age also show higher rates. Combining prior visits and length of stay gives a simple risk grouping: the 683 high-risk patients (about 1% of the data) have a 17.42% rate, compared with 8.14% for the lower-risk group. The results show **associations, not causes**.

Key figures

Measure	Value
Raw records / cleaned patients	101,766 / 69,979
Readmitted within 30 days	6,283 (8.98%)
Prior visits: lowest to highest group	7.94% (0 visits) to 18.00% (6+ visits)
Length of stay: lowest to highest group	7.53% (1-3 days) to 11.79% (8+ days)
Age: lowest to highest group	6.19% (0-29) to 10.62% (80+)
High / Medium / Lower risk	17.42% (683) / 12.13% (13,109) / 8.14% (56,187)
Highest single combination	6+ prior visits and 8+ day stay: 25.19%

1. Business problem

Hospital readmissions within 30 days are costly and are often a sign that a patient's care could be improved. If a hospital knows which patient and stay factors are linked to higher readmission, it knows where to look first when planning discharge and follow-up care. The project answers three business questions:

- What is the overall 30-day readmission rate?
- Which factors (prior visits, length of stay, age, diagnosis, medication change, A1C testing, admission type) are linked to higher readmission?
- Can these factors be combined into a simple risk grouping that shows which patients need extra attention?

2. Dataset

Item	Detail
Name	Diabetes 130-US Hospitals for Years 1999-2008
Source	UCI Machine Learning Repository, licence CC BY 4.0. https://archive.ics.uci.edu/dataset/296/diabetes-130-us-hospitals-for-years-1999-2008
Citation	Clore, J., Cios, K., DeShazo, J., and Strack, B. (2014). DOI: https://doi.org/10.24432/C5230J
Size	101,766 hospital encounters, 50 columns, 130 US hospitals, 10 years
Target	readmitted: '<30' (readmitted within 30 days), '>30' or 'NO'. A flag readmitted_30 (1 or 0) is created for the analysis.
Availability	The raw and cleaned data files are not stored in the GitHub repository. They are downloaded from the source link above.

3. Tools and workflow

Stage	Tool	Output
Profile, clean, validate, create features, charts	Python 3.12 (pandas, matplotlib) in a Jupyter notebook and PyCharm scripts	69,979 patients, 55 columns, 7 charts
Store and analyse	MySQL (database diabetes_db, table diabetes_encounters)	Schema file and 9 SQL queries
Model and visualise	Power BI Desktop (DAX, star schema, slicers)	3-page dashboard
Share	Git and GitHub	Code, notebook, SQL, charts, screenshots and README

Python reads the raw file and writes a cleaned file. A script then loads that file into MySQL. The password is typed when the script runs, so it is never stored in the code. Power BI connects to the MySQL table.

4. Data profiling and cleaning

The profiling step looked at the size of the data, column types, missing values (written as '?') and duplicates before anything was changed. It showed that the same patient can appear in many rows, that the weight column is almost empty, and that race, payer code, medical specialty and the second and third diagnoses have missing values. Each cleaning rule below has a reason, and the row counts come from the cleaning log.

Cleaning rule	Rows removed	Reason
Drop weight column	-	96.9% missing
Drop columns with one value in every row (examide, citoglipton)	-	No information
Remove invalid gender	3	Cannot be used in gender analysis
Remove patients who died or went to hospice	2,423	They cannot be readmitted and would distort the rate
Remove rows with no primary diagnosis	20	Patients are grouped by diagnosis
Keep only the first encounter per patient	29,341	Each patient counts once, so frequent patients do not dominate
Fill other missing values with 'Unknown'; treat 'None' in test columns as 'Not Tested'	-	Avoids guessing values; 'None' means the test was not done
Result	**31,787 in total**	**101,766 records reduced to 69,979 patients, 47 columns, 0 missing values**

The rule that keeps one encounter per patient removed 92% of all removed rows. These were repeat visits by the same patients, not bad data.

Validation

- Every patient appears only once (checked in the notebook and again in SQL).
- No missing values remain.
- No invalid gender and no died or hospice patients remain.
- The target column only has the values '<30', '>30' and 'NO'.

5. Feature engineering

New column	How it is made
readmitted_30	1 if readmitted within 30 days, otherwise 0 (the target)
age_group	10-year bands grouped into 0-29, 30-59, 60-79 and 80+
stay_group	Short (1-3 days), Medium (4-7 days), Long (8+ days)
prior_visits	Outpatient + emergency + inpatient visits in the previous year
meds_active, meds_changed	Number of diabetes medicines taken, and number changed up or down
diag_group	Primary diagnosis grouped from ICD-9 codes (Circulatory, Respiratory, Injury and others)
admission_type	Readable name for the admission type code (Emergency, Urgent, Elective, Other/Unknown)

6. Exploratory analysis

Each chart shows the 30-day readmission rate for the groups of one factor. The red dashed line is the overall rate of 8.98%. A bar above the line means that group has a higher readmission rate than average.

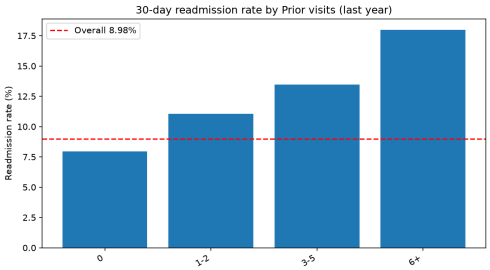


Figure 1: Prior visits. The strongest factor (7.94% to 18.00%).

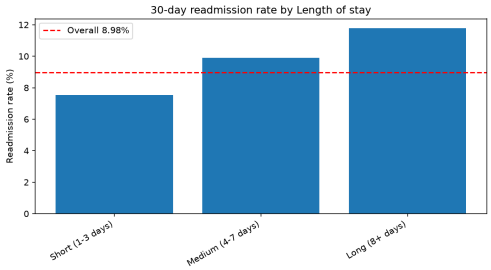


Figure 2: Length of stay (7.53% to 11.79%).

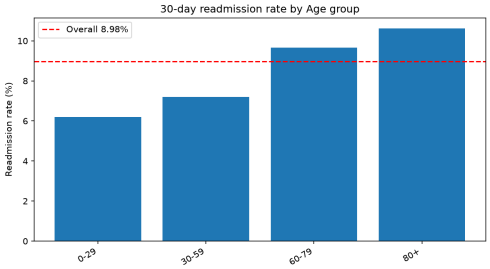


Figure 3: Age group (6.19% to 10.62%).

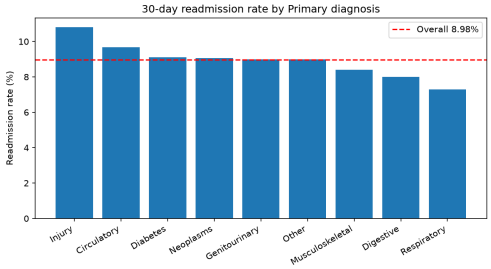


Figure 4: Primary diagnosis (Respiratory 7.30% to Injury 10.80%).

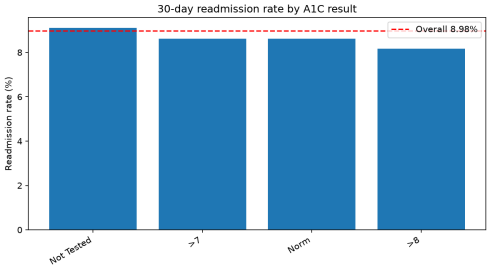


Figure 5: A1C result. 81.65% of patients were not tested.

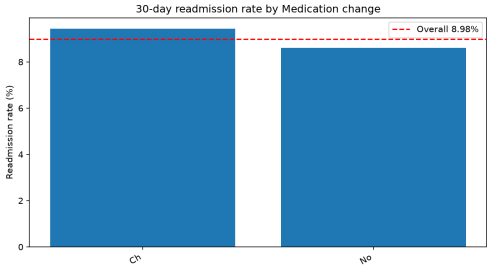


Figure 6: Medication change (Ch = changed): 9.44% vs 8.60%.

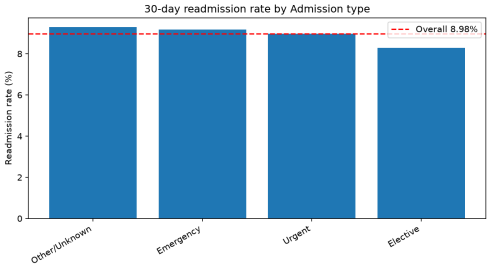


Figure 7: Admission type (8.29% to 9.29%), a weak factor.

What the charts show

- **Prior visits** is the strongest factor. The rate rises steadily from 7.94% with no prior visits to 18.00% with six or more.
- **Length of stay** and **age** also rise clearly: from 7.53% to 11.79% by stay, and from 6.19% to 10.62% by age.
- **Primary diagnosis** matters: Injury (10.80%) is highest and Respiratory (7.30%) is lowest.
- **Medication change** and **admission type** show differences of less than one percentage point.
- **A1C:** 81.65% of patients had no A1C test, and this group had the highest rate (9.11%), so conclusions about A1C are limited.

7. SQL analysis in MySQL

The cleaned data was loaded into the table **diabetes_encounters** (69,979 rows, 55 columns, primary key encounter_id). The file **00_schema.sql** creates the database and table. The file **06_sql_analysis.sql** starts with three data checks (row count, one row per patient, target column values 0 and 1) and then runs 9 analysis queries. Each query groups the patients and uses AVG(readmitted_30) x 100 as the readmission rate.

#	Business question	Result
1	Rate by age group	6.19% (0-29) rising to 10.62% (80+)
2	Rate by number of prior visits	7.94% (0 visits) rising to 18.00% (6+)
3	Rate by length of stay	7.53% (1-3 days) rising to 11.79% (8+ days)
4	Rate by medication change	9.44% (changed) vs 8.60% (no change)
5	Rate by primary diagnosis group	Injury 10.80% highest, Respiratory 7.30% lowest
6	Rate by A1C result and share of patients	81.65% not tested; this group has the highest rate (9.11%)
7	Rate by combined risk segment	High 17.42% (683), Medium 12.13% (13,109), Lower 8.14% (56,187)
8	Overall summary (KPIs)	69,979 patients, 6,283 readmitted, 8.98%
9	Rate by admission type	8.29% (Elective) to 9.29% (Other/Unknown)

Example: the risk segment query (Query 7)

```
CASE WHEN prior_visits >= 3 AND time_in_hospital >= 8 THEN 'High risk'
WHEN prior_visits >= 3 OR time_in_hospital >= 8 THEN 'Medium risk' ELSE 'Lower risk' END
```

The risk segments are a rule-based grouping built from the two strongest factors. They are not a prediction model.

8. Data model and DAX

The Power BI model is a **star schema**. The table diabetes_encounters is the central fact table (one row per patient). Four small lookup (dimension) tables are connected to it with one-to-many relationships, and they feed the slicers on every page.

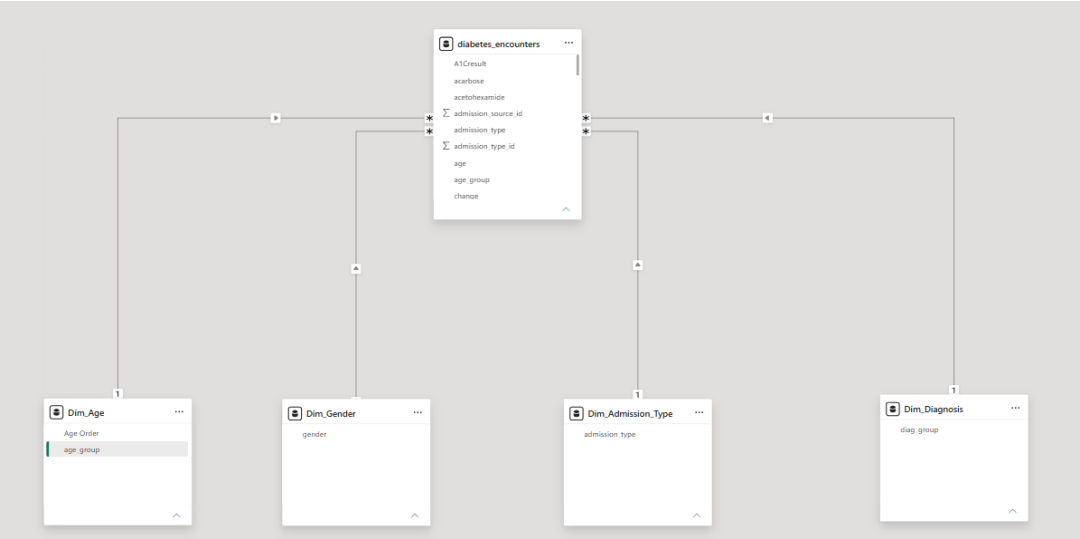


Figure 8: Power BI data model (star schema).

Lookup table	Column	Connected to diabetes_encounters column	Relationship
Dim_Age (sorted by Age Order)	age_group	age_group	One to many, single direction
Dim_Gender	gender	gender	One to many, single direction

Lookup table	Column	Connected to diabetes_encounters column	Relationship
Dim_Admission_Type	admission_type	admission_type	One to many, single direction
Dim_Diagnosis	diag_group	diag_group	One to many, single direction

DAX measures and calculated column

```
Total Patients = COUNTROWS(diabetes_encounters)
Readmitted Patients = SUM(diabetes_encounters[readmitted_30])
Readmission Rate = DIVIDE([Readmitted Patients], [Total Patients])
Risk Segment = IF(AND(prior_visits >= 3, time_in_hospital >= 8), "1 - High risk",
IF(OR(prior_visits >= 3, time_in_hospital >= 8), "2 - Medium risk", "3 - Lower risk"))
```

Helper columns (Stay Order, Prior Visits Group, Prior Visits Order, Medication Change) sort and label the groups. DIVIDE is used so that an empty filter returns a blank instead of an error. The risk labels are shortened in this report.

Verification of the calculations

The same numbers were produced in three tools and compared. Slicer tests prove that the relationships filter correctly.

Measure	Python notebook	SQL	Power BI
Total patients	69,979	69,979	69,979
Readmitted within 30 days	6,283	6,283	6,283
Overall rate	8.98%	8.98%	8.98%
High risk (patients, rate)	683, 17.42%	683, 17.42%	683, 17.42%
Medium risk (patients, rate)	13,109, 12.13%	13,109, 12.13%	13,109, 12.13%
Lower risk (patients, rate)	56,187, 8.14%	56,187, 8.14%	56,187, 8.14%
Slicer test: Age 80+	10.62%	10.62%	12,869 patients, 1,367 readmitted, 10.62%
Slicer test: Injury	10.80%	10.80%	4,694 patients, 507 readmitted, 10.80%
Slicer test: Emergency	9.18%	9.18%	35,475 patients, 3,256 readmitted, 9.18%

Slicer tests: the readmission rate for each group in the notebook and SQL (charts and Query 5, 1 and 9) equals the Power BI card when that slicer is selected.

9. Power BI dashboard

The dashboard has three pages with a page navigator at the top, four synced slicers (age, gender, admission type, diagnosis) and a Reset all filters button. The three cards (Total Patients, Readmitted within 30 days, 30-Day Readmission Rate) are on every page and react to the slicers.

Page 1: Summary

Shows the overall picture: readmission rate by age group, prior visits, length of stay and primary diagnosis, and the share of patients by A1C result.

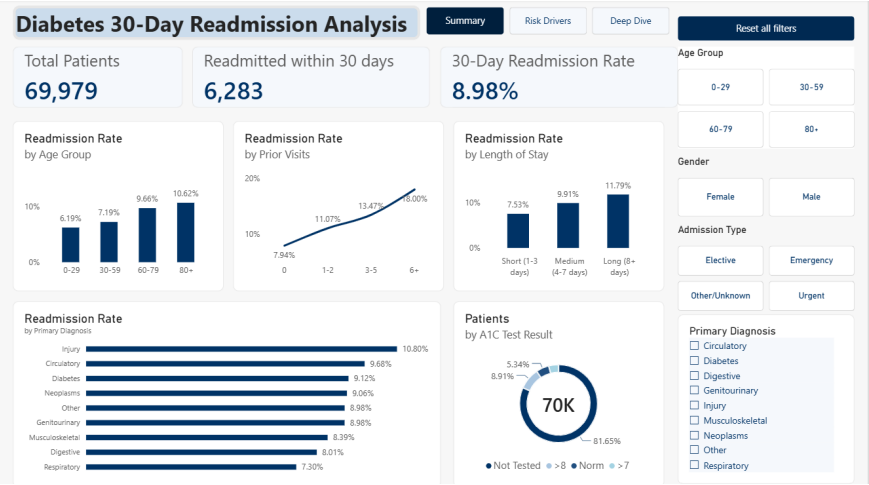


Figure 9: Summary page.

Page 2: Risk Drivers

Shows who is most at risk: a matrix of prior visits against length of stay (highest cell 25.19% for 6+ prior visits and 8+ days), medication change, admission type and the combined risk segments.



Figure 10: Risk Drivers page.

Page 3: Deep Dive

A decomposition tree lets the user split the readmission rate by diagnosis and then by age group. A treemap shows patients by diagnosis, and a text box lists the key insights.

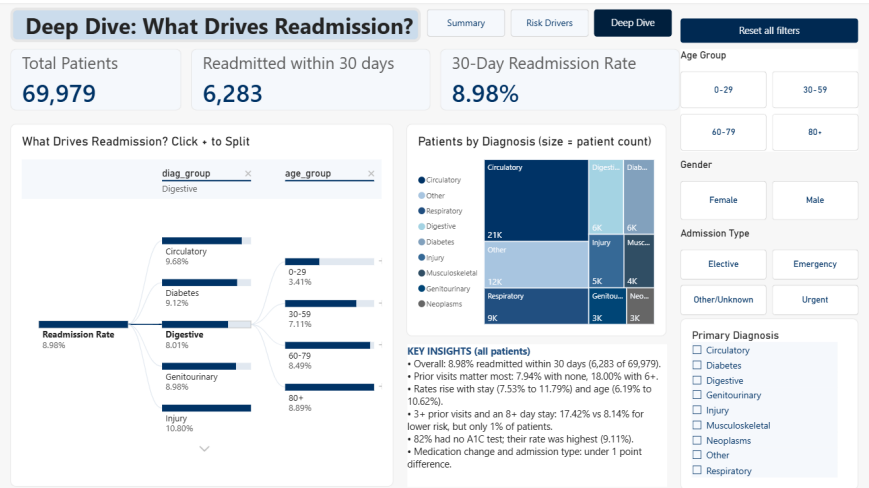


Figure 11: Deep Dive page.

10. Insights and business recommendations

These suggestions come from the patterns above. They are ideas to test, not proven actions.

- **Target follow-up at the high-risk group.** Patients with 3+ prior visits and an 8+ day stay have a 17.42% rate, about twice the overall 8.98%, and they are only 683 patients (about 1%). A small, focused effort such as extra discharge planning or a follow-up call can reach them at low cost.
- **Use prior visits as an early warning flag at admission.** The rate rises from 7.94% to 18.00%, so care teams can flag these patients on the first day.
- **Review discharge planning after long stays.** Stays of 8+ days have an 11.79% rate compared with 7.53% for 1-3 days.
- **Give extra attention to older patients and injury-related admissions.** Patients aged 80+ (10.62%) and Injury diagnoses (10.80%) are above the overall rate.
- **Record A1C results consistently.** 81.65% have no A1C result, so the data cannot show whether testing matters.
- **Do not rely on admission type or medication change alone.** Both show small differences.

11. Limitations

- The analysis is descriptive: it shows associations, not causes.
- The risk segments are simple rules, not a validated prediction model, and must not be used for clinical decisions.
- The high-risk group is small (683 patients), so its rate is based on a small group.
- 81.65% of patients have no A1C result, so conclusions about A1C are limited.
- Only each patient's first encounter is kept (29,341 repeat encounters removed), so the results describe first stays.
- The data covers 1999-2008 in US hospitals, so patterns may differ today or in other places.

12. Conclusion

The project took a raw hospital dataset through profiling, cleaning, validation, feature engineering, SQL analysis and a star-schema Power BI dashboard, and checked the same numbers in three tools. Prior visits, length of stay and age are the clearest factors linked to 30-day readmission, and a simple two-factor grouping separates a small high-risk group (17.42%) from the lower-risk group (8.14%). The next step is to test these ideas with a predictive model validated on separate data.

Project files

GitHub repository: https://github.com/anumasi-analytics/diabetes-readmission-analysis
notebooks/diabetes_readmission_analysis.ipynb | python/01 to 05 | sql/00_schema.sql and sql/06_sql_analysis.sql | charts/ | powerbi/screenshots/ | README.md
Power BI file: diabetes_hospital_project_v2.pbix (submitted separately).